

JKBOSE 2026

CLASS 11TH

#XUV



CHEMICAL THERMODYNAMICS

LECTURE-1

CHEMISTRY

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ON
TOP.



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Topics To Covered

- 1** Concept of System & Surrounding
- 2** Extensive & Intensive Properties

REVISION



MOD

i) For molecules upto N_2 ($Z \leq 7$): $\rightarrow \sigma_{1s} < \sigma_{1s}^* < \sigma_{2s} < \sigma_{2s}^* < \pi_{2px} = \pi_{2py} < \sigma_{2pz} < \pi_{2px}^* = \pi_{2py}^* < \sigma_{2pz}^*$

ii) For molecules. $Z > 7$ (O_2, F_2, Ne_2): $\rightarrow \sigma_{1s} < \sigma_{1s}^* < \sigma_{2s} < \sigma_{2s}^* < \sigma_{2pz} < \pi_{2px} = \pi_{2py} < \pi_{2px}^* = \pi_{2py}^* < \sigma_{2pz}^*$

i) $H_2, He_2, Li_2, Be_2,$

B_2, C_2 ✓

N_2, O_2, F_2, Ne_2

5) B_2

Atomic orbital $\bar{e}$ configuration

$$B(Z=5) = 1s^2 2s^2 2p^1$$

Total electron = $10\bar{e}$

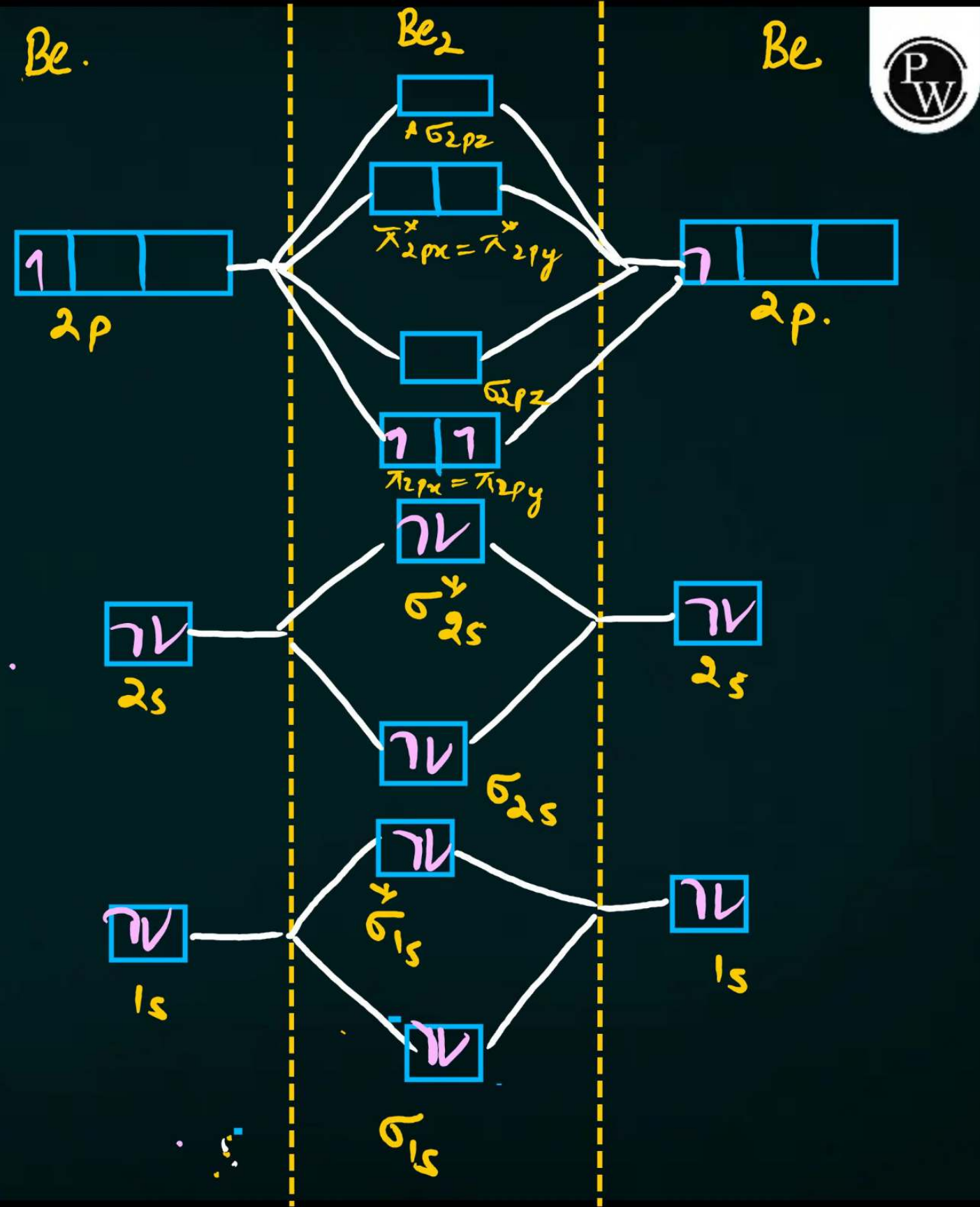
Molecular orbital electronic configuration

$$\sigma_{1s}^2 < \sigma_{1s}^{*2} < \sigma_{2s}^2 < \sigma_{2s}^{*2} < \pi_{2px}^1 = \pi_{2py}^1$$

• Stability = $N_b > N_a$ i.e. stable.

$$BO = \frac{1}{2} (N_b - N_a) = \frac{1}{2} (8 - 4) = \frac{2}{2} = 1$$

• Magnetic behaviour $\rightarrow$ paramagnetic.



7) N_2

AO electronic configuration of N ($Z=7$).
 $1s^2 2s^2 2p^3$

Total no. of $\bar{e} = 14 \bar{e}$

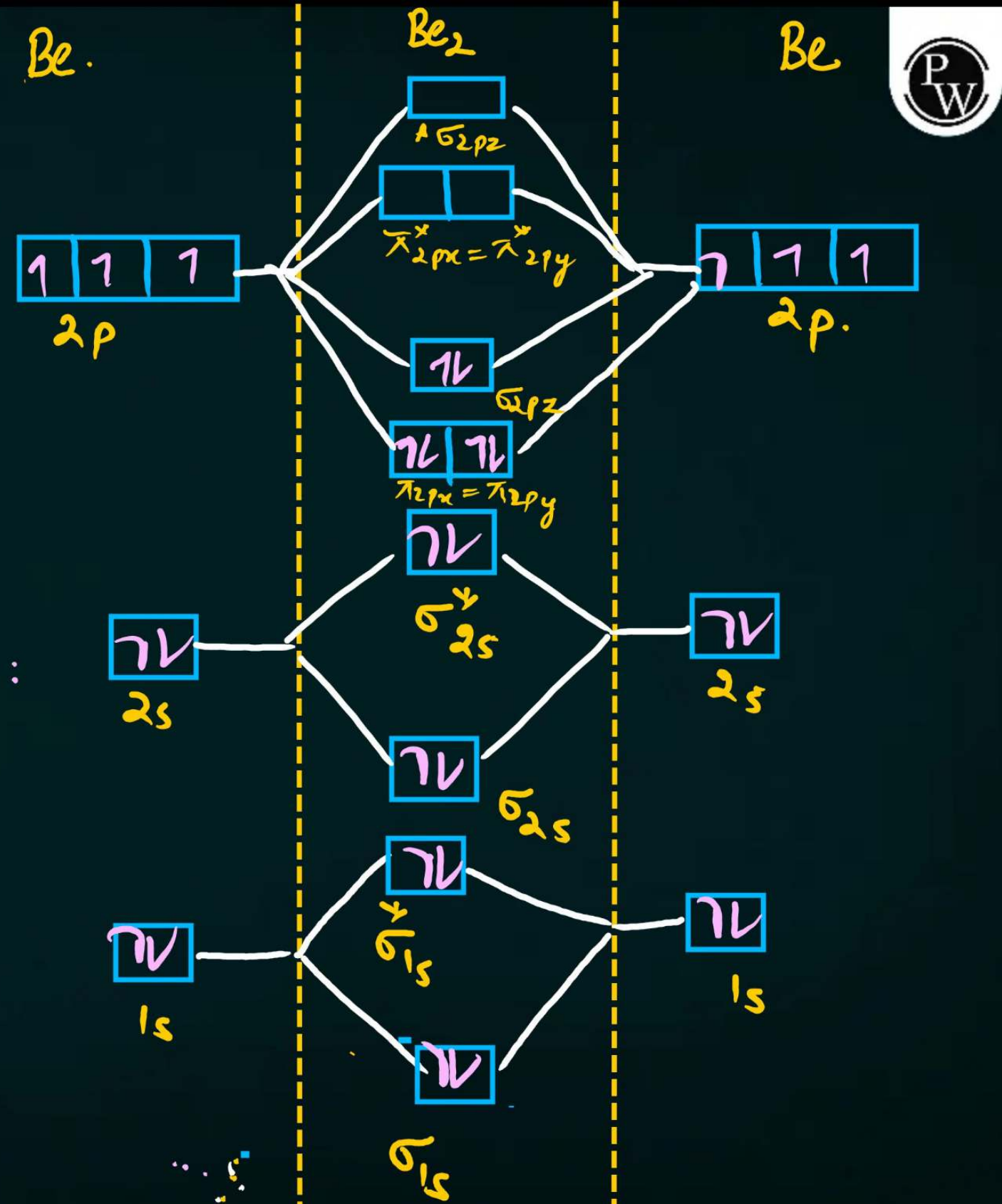
• MO electronic configuration $\rightarrow$

$$\sigma_{1s}^2 < \sigma_{1s}^{*2} < \sigma_{2s}^2 < \sigma_{2s}^{*2} < \pi_{2px}^2 = \pi_{2py}^2 < \sigma_{2pz}^2$$

• Stability $\rightarrow N_b > N_a$ i.e. stable

$$\bullet BO = \frac{1}{2}(N_b - N_a) = \frac{1}{2}(10 - 4) = \frac{6}{2} = 3$$

• Magnetic behaviour $\rightarrow$ Diamagnetic.



8) O_2

AO electronic configuration of $O(z=8)$.
 $1s^2 2s^2 2p^4$

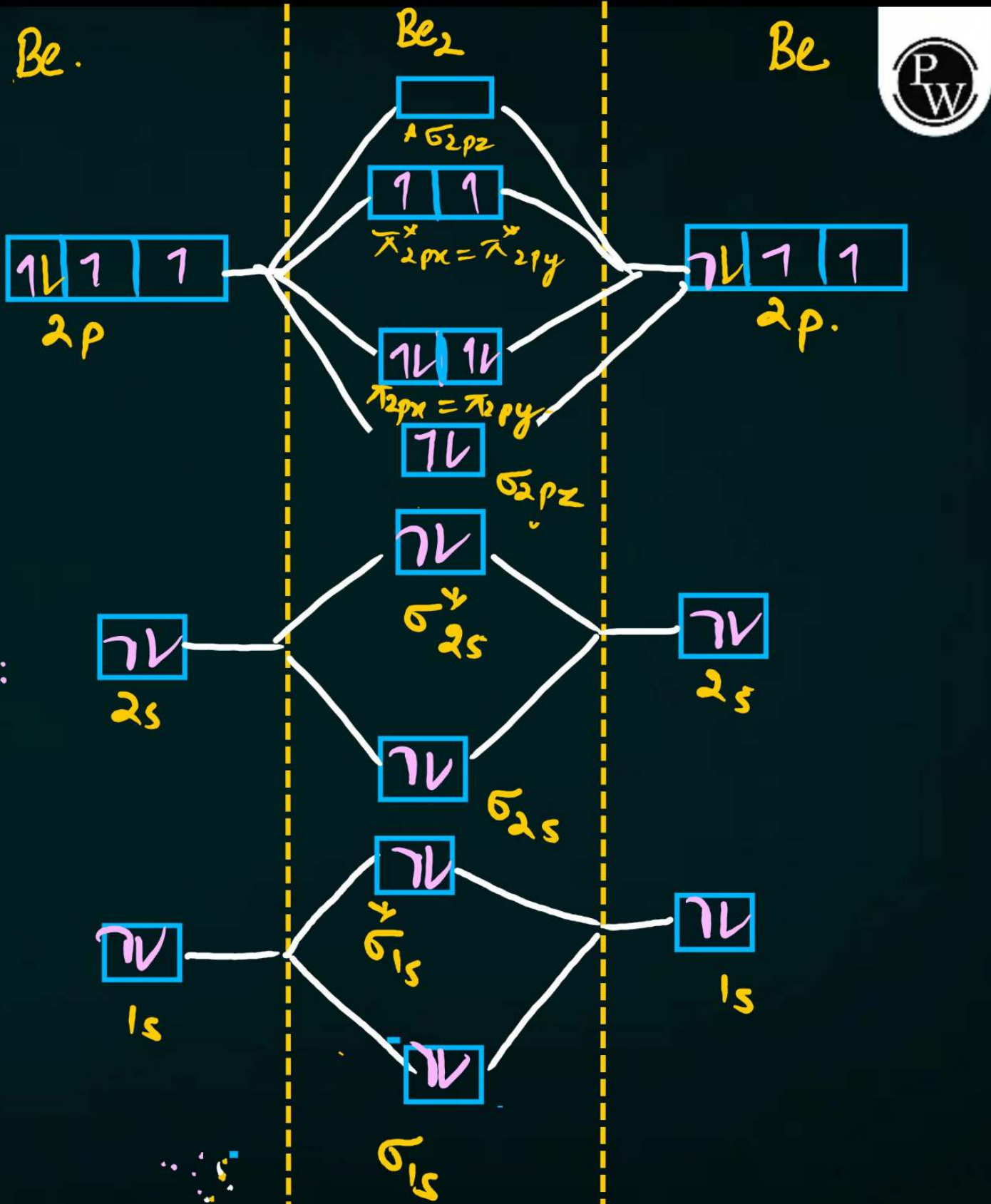
Total no. of $e^- = 16e^-$

• MO electronic configuration $\rightarrow \sigma_{1s}^2 < \sigma_{1s}^{*2} < \sigma_{2s}^2 < \sigma_{2s}^{*2} < \sigma_{2p_z}^2 < \pi_{2p_x}^2 = \pi_{2p_y}^2 < \pi_{2p_x}^{*1} = \pi_{2p_y}^{*1}$

• Stability $\rightarrow N_b > N_a$ i.e. stable.

• $BO = \frac{1}{2}(N_b - N_a) = \frac{1}{2}(10 - 6) = \frac{1}{2}(4) = 2$

• Magnetic behaviour $\rightarrow$ paramagnetic.

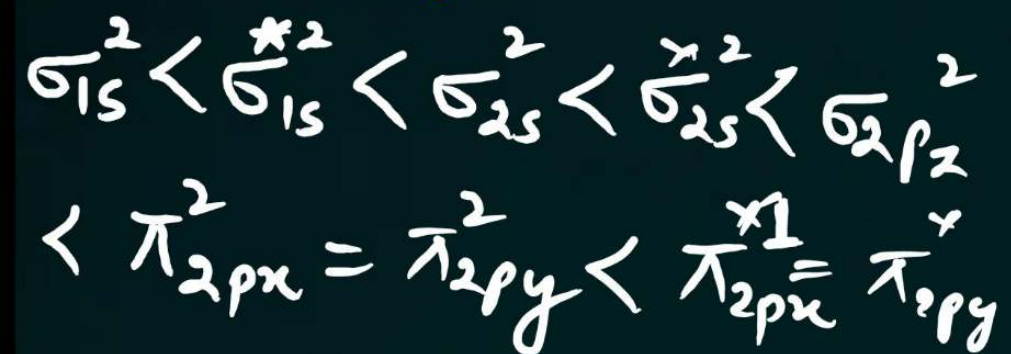


Home Work.





MO e^- Configuration

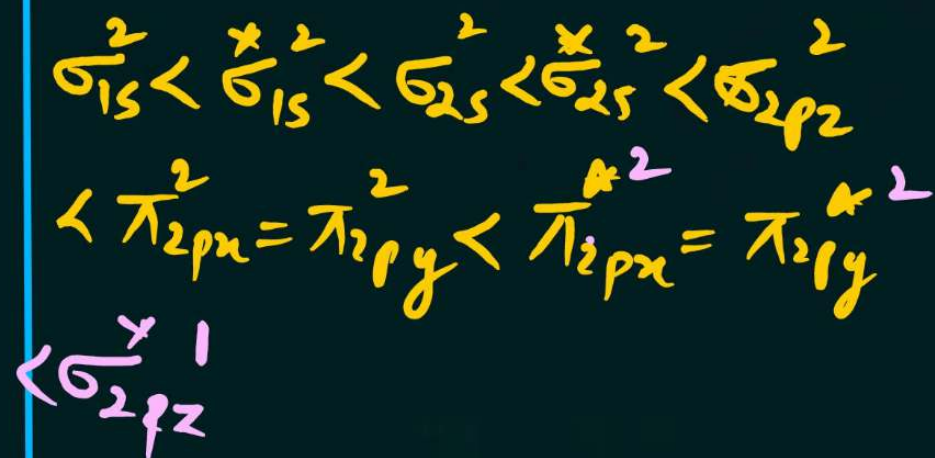
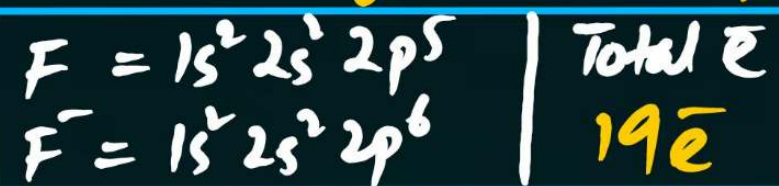


Stability: $N_b > N_a$ i.e stable

$$B.O = \frac{1}{2}(N_b - N_a) = \frac{1}{2}(10 - 5) = \frac{5}{2}$$

$$= 2.5.$$

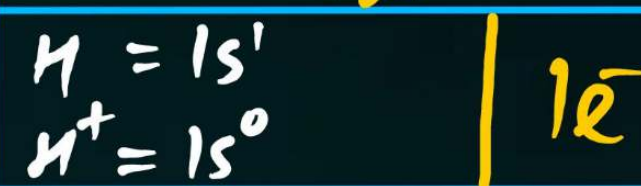
Magnetic behaviour = Paramagnetic



$N_b > N_a$ i.e stable.

$$\frac{1}{2}(10 - 9) = \frac{1}{2} = 0.5$$

Paramagnetic



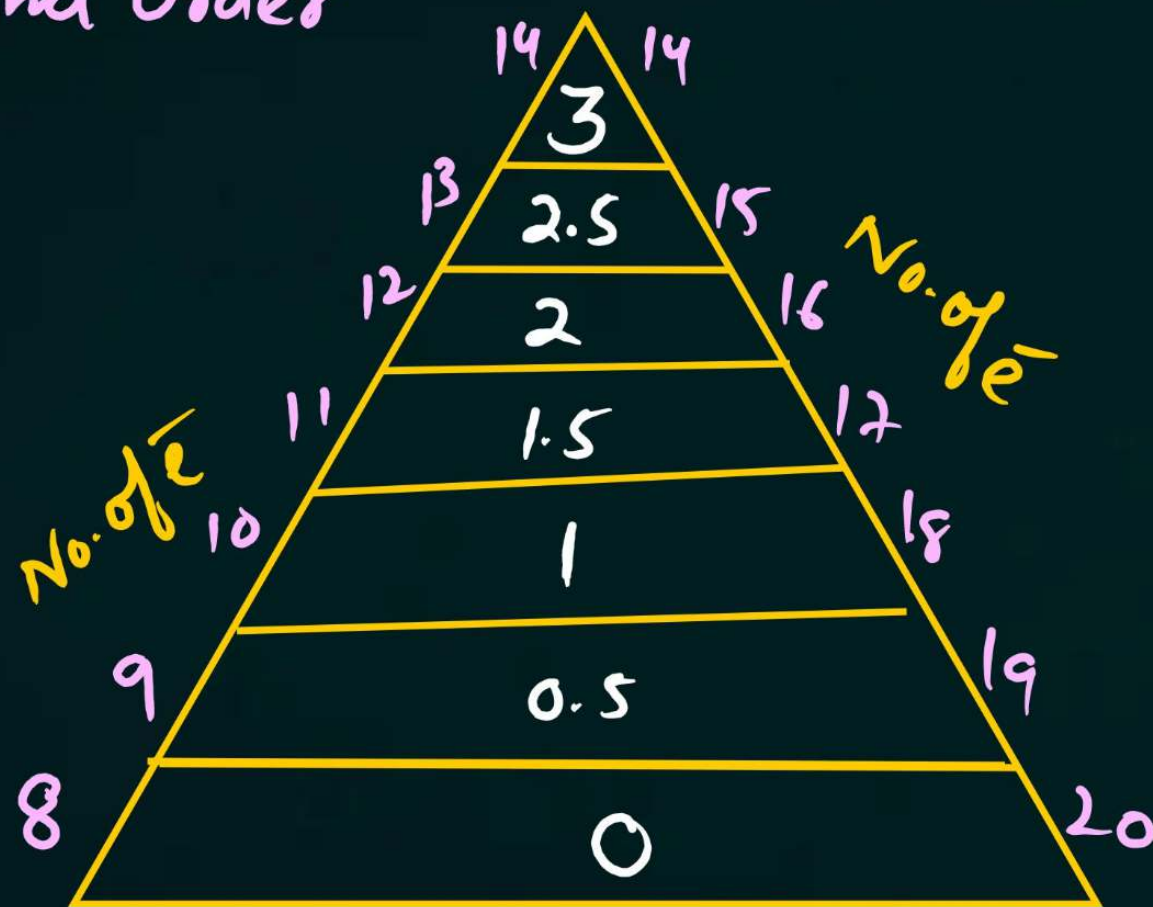
$N_b > N_a$ i.e stable.

$$\frac{1}{2}(1 - 0) = \frac{1}{2} = 0.5$$

Paramagnetic.

BOT TRICK.

1) For Bond Order



2) Magnetic behaviour

i) If total no. of e^- are odd = always paramagnetic

ii) If total no. of e^- are even = all diamagnetic except (10 e^- & 16 e^-)

$\downarrow$ $\downarrow$
 B_2 O_2

Q. What is the bond order of molecule O_2^-

a) 2

b) 1.5

c) 2.5

d) 3

$$\text{Total no. of } e^- = 2 \times 8 + 1 = 17e^-$$

12	13	14	15	16	17
2	2.5	3	2.5	2	1.5

Q. Predict the BO & magnetic behaviour of F_2^-

a) 1.5, dia ✗

b) 2.5, para ✗

c) 0.5, para

d) 0.5, dia

Total no. of $e^- = ?$

$$= 2 \times 9 + 1$$

$$= 19e^-$$

14	15	16	17	18	19
3	2.5	2	1.5	1	0.5

- 1) Sbcoc
 - 2) Str. of atom
 - 3) Classification
 - 4) Chemical bonding
 - 5) Thermodynamics
- $\left. \begin{array}{l} 1) \\ 2) \end{array} \right\} \text{PC}$
 $\left. \begin{array}{l} 3) \\ 4) \end{array} \right\} \text{IOE}$
 $\left. \begin{array}{l} 5) \end{array} \right\} \text{PC.}$

MUBARAK.

UNIT-V: THERMODYNAMICS

JKBOSE ↓

09 MARKS

- Concepts of system, types of systems, surrounding, work, heat; energy intensive and extensive properties, state functions. First Law of Thermodynamics, internal energy, enthalpy, heat capacity, specific heat, molar heat capacity, measurement of E and H, Hess's law of constant heat summation, enthalpy of bond dissociation, combustion; formation, atomization, sublimation, phase transition ionization and dilution.
- Introduction of entropy as a state function, free energy change for spontaneous and non-spontaneous process and equilibrium.

THERMODYNAMICS. It is the branch of Science that deals with the study of various types of energies and their interconversion.

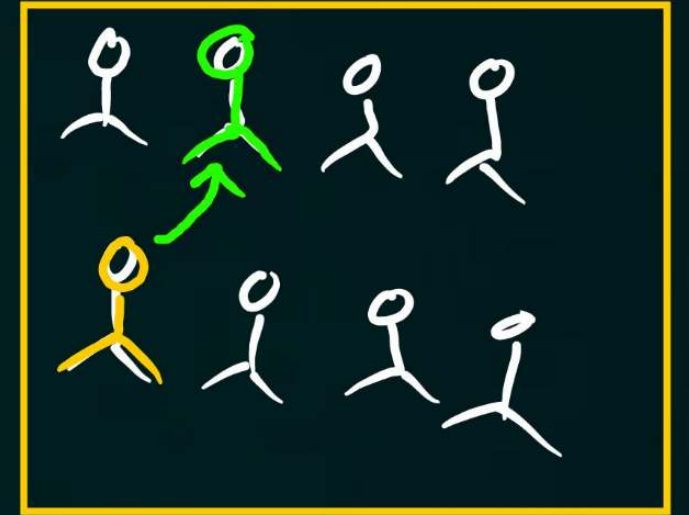
Concept of System & Surrounding



SYSTEM $\rightarrow$ Part of the Universe in which we are interested is called system.

SURROUNDING $\rightarrow$ Part of universe except system is called surrounding.

Universe is comprised of System & Surrounding.
i.e. $\text{Universe} = \text{System} + \text{Surrounding}$.



TYPES OF SYSTEMS.

1) OPEN SYSTEM $\rightarrow$ System that exchanges energy as well as matter with the surrounding. e.g. Water placed in an open vessel.

2) CLOSED SYSTEM $\rightarrow$ System that exchanges only energy with surrounding but not the matter. e.g. Water placed in a closed vessel.

3) ISOLATED SYSTEM $\rightarrow$

System which neither exchanges energy nor the matter with the surrounding. e.g. Water placed in an thermos.

INTENSIVE AND EXTENSIVE PROPERTIES.

a) Intensive Properties $\rightarrow$ Properties of a system that are independent of amount of substance.

e.g Density, Viscosity, Refractive index
Pressure, Temperature, concentration etc.

b) EXTENSIVE PROPERTY.

Properties of a system that depends upon the amount of substance.

e.g mass, volume, moles,
Enthalpy, Internal energy,
Heat Capacity etc

- Ratio of two extensive properties is always equal to intensive property

e.g $\text{Density} = \frac{\text{Mass}}{\text{Volume}}$

QUESTION



3M

What are intensive and extensive properties? Give two examples of each.

[2025]

QUESTION

What do you understand by open and isolated systems? Give examples.

③ 12 ✓
[2025]



QUESTION

Which of the following is an extensive property?

- 1 Temperature
- 2 Density
- 3 Gibbs free energy
- 4 Molar volume

QUESTION

Which of the following statement is correct?

- 1 The presence of reacting species in a covered beaker is an example of open system.
- 2 There is an exchange of energy as well as matter between the system and the surroundings in a closed system.
- 3 The presence of reactants in a closed vessel made up of copper is an example of a closed system.
- 4 The presence of reactants in a thermos flask or any other closed insulated vessel is an example of a closed system.

QUESTION

Which of the following is an intensive property?

[PYQ]

1 Heat capacity

2 Mass

3 Density

4 Volume

REVISION

Thank
You